

so we can get

$$\begin{cases} \frac{dx_1}{dt} = x_2 + \frac{b_{11}}{b_{10}}x_1x_2 + \frac{b_{20}}{b_{10}^2}x_2^2 + \frac{b_{12}}{b_{10}}x_1^2x_2 + \frac{b_{21}}{b_{10}^2}x_1x_2^2 + \frac{b_{30}}{b_{10}^3}x_1^3 + O(\|x\|^4) \\ \frac{dx_2}{dt} = a_{11}x_1x_2 + \frac{a_{20}}{b_{10}}x_2^2 + \frac{a_{21}}{b_{10}}x_1x_2^2 + a_{12}x_1^2x_2 + \frac{a_{30}}{b_{10}^2}x_2^3 + O(\|x\|^4) \end{cases}$$

$$\begin{cases} x_3 = x_1 \\ x_4 = x_2 + \frac{b_{11}}{b_{10}}x_1x_2 + \frac{b_{20}}{b_{10}^2}x_2^2 + \frac{b_{12}}{b_{10}}x_1^2x_2 + \frac{b_{21}}{b_{10}^2}x_1x_2^2 + \frac{b_{30}}{b_{10}^3}x_1^3 + O(\|x\|^4) \end{cases}$$

$$\begin{cases} x_1 = x_3 \\ x_2 = x_4 + v_{11}x_3x_4 + v_{02}x_4^2 + v_{21}x_3^2x_4 + v_{12}x_3x_4^2 + v_{03}x_4^3 + O(\|x\|^4) \end{cases}$$

wherein:

$$\begin{aligned} v_{11} &= -\frac{b_{11}}{b_{10}}, v_{02} = -\frac{b_{20}}{b_{10}^2}, \\ v_{21} &= \frac{b_{11}^2 - b_{12}b_{10}}{b_{10}^2}. \\ v_{12} &= \frac{3b_{11}b_{20} - b_{21}b_{10}}{b_{10}^3}, v_{03} = \frac{2b_{20}^2}{b_{10}^4}. \end{aligned}$$

Substitute the new variables (x_3, x_4) into the original system, and we can get:

$$\begin{cases} \frac{dx_3}{dt} = x_4 \\ \frac{dx_4}{dt} = e_{11}x_3x_4 + e_{02}x_4^2 + e_{21}x_3^2x_4 + e_{12}x_3x_4^2 + e_{03}x_4^3 + O(\|x\|^4) \end{cases} \quad (14)$$

wherein:

$$\begin{aligned} e_{11} &= a_{11}, \quad e_{02} = \frac{a_{20} + b_{21}}{b_{10}}, \\ e_{21} &= a_{12} + \frac{a_{11}b_{11}}{b_{10}} + a_{11}v_{11}, \\ e_{12} &= a_{12}v_{02} + \frac{(a_{20} + b_{11})b_{10} + a_{21}b_{10} + 2a_{11}b_{20} + a_{20}b_{11} + b_{11}^2 + 2b_{10}b_{12}}{b_{10}^2}, \\ e_{03} &= \frac{2(a_{20} + b_{11})b_{10}^2v_{02} + a_{30} + b_{21} + 2a_{20}b_{20} + b_{11}b_{20}}{b_{10}^3}. \end{aligned}$$

Then make the following transformation for the system (14):

$$\begin{cases} x_5 = x_3 \\ x_6 = x_4 - e_{02}x_3x_4 \end{cases}$$